

DevOps – Lab Program-3

Documentation

Title: Code a Dockerized python flask or node js application

Software Requirements:

- Docker
- Node.js
- Express.js
- VS Code / Terminal

Files and Folder Structure:

Prg3/

```
|
|— node_modules/
|— index.js
|— package.json
|— package-lock.json
|— Dockerfile
```

Procedure

1. Open the project folder on your computer.
2. Install all required project files (dependencies).
3. Start the server for the project.
4. Open your web browser.
5. In the address bar, type “localhost:7171”.
6. The browser will display the message “Hello World!”.

Source Code and their Snapshots

[Index.js](#)

```
const express = require('express');
const app = express();
const port = 7001;

app.get('/', (req, res) => {
  res.send('Hello FOR PROGRAM 3');
});

app.listen(port, () => {
  console.log(`Server is running at http://localhost:${port}`);
});
```

Package.json

```
{
  "name": "p3",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node index.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  }
}
```

Dockerfile

```
# Specifying FROM
FROM node:20-alpine

# specifying working directory
WORKDIR /app

# Copying necessary files
COPY package*.json ./

# Installing dependencies
RUN npm install

# Copying again
COPY . .

# Exposing the port
EXPOSE 7001npm

# Start the app
CMD ["npm", "start"]
```

OUTPUT

